Certified ceilings for a Mertens-type extremal linear program, with exact dual certificates at reaches 4800 and 12000

Kevin Russell*

ProjectForty2 — CHRONOS agent

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Abstract

For $K \in \mathbb{N}$ and tolerance $\tau = 1.0001$, let

$$S^*(K) \;=\; \max\Big\{ -\sum_{k=2}^{K} f(k)\,\frac{\ln k}{k} \;:\; f \in [-10,10]^{K-1}, \\ \sum_{k=2}^{K} f(k)\Big(\Big\lfloor \frac{m}{k} \Big\rfloor - \frac{m}{k}\Big) \le \tau \text{ for } m = 1,\dots,10K-1 \Big\},$$

the extremal value of the scoring rule of the EinsteinArena "prime-number-theorem" benchmark board. We prove

$$S^*(4800) \;\le\; 0.9963688817172828744325041, \quad S^*(12000) \;\le\; 0.9974876103072528157057480,$$

by weak-duality *box certificates*: explicit dyadic-rational dual vectors $y = Y/2^{48}$, entrywise nonnegative, whose certified bound $B(y) = \tau \sum_m y_m + 10 \sum_k |r_k|$ is evaluated with exact integer residual accumulation and outward-rounded 400-bit interval enclosures of $\ln k/k$; no unenclosed floating-point scalar enters the certified inequality — every transcendental quantity is replaced by an outward-rounded high-precision interval, and the log enclosures are themselves cross-audited (dual-precision nesting at 400 vs 800 bits over every k, plus a rational atanh-series bracket independent of any log oracle) — and an independent skeptic harness (fresh residual computation by a different route, pure-rational spot checks, hash pinning) passes all checks on both certificates. Numerically, the certificates are close to the floating-point LP optima, which sit 5.6×10^{-7} (reach 4800) and 2.7×10^{-5} (reach 12000) below the certified ceilings (these optima are float measurements, not a certified lower bracket). The problem is a finite Mertens-type program: the Möbius function is its infinite-reach extremal shadow — μ satisfies the constraint with $E(m) \equiv 1$ and objective 1, both classical equivalents of the prime number theorem — but no finite truncation of μ is feasible (measured violation 104.2 at reach 48000), which is why the board's leaders are sparse feasible surrogates rather than truncated Möbius. Alongside the two theorems we report measurements (floats, clearly labeled, none of them certificates): the reach-12000 optimal support is dense (10255 of 11999 keys), a cardinality-penalty curve quantifying the board's 2000-key cap, and a certified bracket for the structure-constant

*Correspondence: `kevinrussell@gmail.com`. Computer-assisted note, prepared with CHRONOS, ProjectForty2's autonomous research agent; the author takes responsibility for all claims. The *constructions* and leaderboard measurements discussed in Section 6 are due to the agents "Agent-Knowledge-Cycle" and "JSAgent" on the EinsteinArena platform [1, 4]; see the attribution statement in Section 1. Verification artifacts are listed in Appendix A.

question raised by the competition agents — the certified floor $c^*(4800) \approx 0.0391$ shows their family's plateau constant $c \approx 0.036$ is unattainable at reach 4800, while at reach 12000 the floor drops to 0.02938 and the best measured 2000-key support reaches only $c \approx 0.0413$, so the capped question stays open in both directions. Items not machine-verified at the time of writing, including the still-running reach-24000 certificate, are listed in a pending-verification ledger. A ceiling is not a construction: nothing on the leaderboard moves because of this note.

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1 Introduction

This note certifies *ceilings* — rigorous upper bounds on the best possible value — for a family of finite linear programs whose origin is a live optimization benchmark. On the EinsteinArena platform, the board `prime-number-theorem` asks agents to submit a partial function $f\colon \{2,\dots\} \to [-10, 10]$ (at most 2000 keys) and scores it by $-\sum_k f(k) \ln k/k$, subject to a verifier constraint recalled in Section 2; the top of the leaderboard has moved $0.99632 \to 0.99652 \to 0.99715 \to 0.99735 \to 0.99740$ over recent weeks as agents push the *reach* (the largest key used) from 16000 to 63998. All leaderboard values are floating-point scores of explicit constructions; none of them, and no value previously discussed on the board, came with a proof of how high the score *can* go at a given reach. The two theorems below supply exactly that missing object at reaches 4800 and 12000: certified, exact-arithmetic upper bounds on the full program value $S^*(K)$, valid for every legal submission at those reaches, sitting 5.6×10^{-7} and 2.7×10^{-5} above the respective LP optima.

Why "prime-number-theorem". The program is a finite, box-constrained relative of Mertens-type sums. As Section 3 recalls, the Möbius function satisfies the constraint *exactly* ($E(m) \equiv 1$ for every m, by an elementary identity), and its objective value is 1 — the statements $\sum_k \mu(k)/k = 0$ and $-\sum_k \mu(k) \ln k/k = 1$ being classical equivalents of the prime number theorem. So the board's score has a natural asymptote at the tolerance τ, approached by $\tau\mu$; but no finite truncation of μ is feasible (we measure a constraint violation of 104.2 at reach 48000), and the interesting mathematics of the competition is which *sparse feasible surrogates* for μ score best under a 2000-key budget.

What this note proves. Theorems 5 and 6 are weak-duality certificates in the sense of Proposition 4: explicit nonnegative dyadic-rational dual vectors $y = Y/2^{48}$ whose certified bound $B(y) = \tau \sum_m y_m + 10 \sum_k |r_k|$ dominates $S^*(K)$. The certified path is exact end to end — integer residual accumulation via $\sum_m Y_m\,(m \bmod k)$, outward-rounded interval enclosures of $\ln k/k$ at 400 bits, directed decimal round-up — and each certificate was re-verified by an independent skeptic harness (Section 4). Everything else in the note — the support density, the cardinality-penalty curve, the structure-constant bracket of Section 6 — is measurement: floating-point values of explicit solves, clearly labeled, confirmed against the platform's own verifier where a legal submission exists, and never called a theorem.

Attribution. This is a certification exercise on a live competition, and credit belongs where the mathematics originated. The multiscale support family driving the current leaderboard (a dense squarefree prefix up to 1800 plus a geometric squarefree tail), the gap decomposition $1 - S \approx c/\ln(10 \cdot \text{reach})$, and an honest reach ladder for it were published on the board by the agent "Agent-Knowledge-Cycle" [1]; the reach-48000 and reach-64000 extensions that hold the lead, the measurement that the family's structure constant has plateaued at $c \approx 0.0360$–0.0361, and the question of whether *any* 2000-key family goes below 0.036 are due to the agent "JSAgent" [4]. Our contribution is the two certified ceilings, the certificate and skeptic machinery, and the measurements of Section 6; the certificate method continues the exact-arithmetic program of our earlier notes on the Erdős minimum overlap constant [5] and on autoconvolution inequalities [6]. This note itself was drafted and verified computer-assisted, by CHRONOS, ProjectForty2's autonomous research agent, operating under the author's direction; the author reviewed the mathematics and takes full responsibility for all claims.

2 The program and the deployed verifier

Fix the tolerance $\tau = 1.0001$ and a reach $K \in \mathbb{N}$. For $f = (f(2), \dots, f(K)) \in [-10, 10]^{K-1}$ define

$$E_f(m) \;=\; \sum_{k=2}^{K} f(k)\Big(\Big\lfloor \frac{m}{k} \Big\rfloor - \frac{m}{k}\Big), \qquad m \in \mathbb{N}, \tag{1}$$

and the program

$$S^*(K) \;=\; \max\Big\{-\sum_{k=2}^{K} f(k)\,\frac{\ln k}{k} \;:\; E_f(m) \le \tau \text{ for all } m \in \{1, \dots, 10K-1\},\; f \in [-10, 10]^{K-1}\Big\}. \tag{2}$$

The feasible set is a nonempty compact polytope ($f \equiv 0$ is feasible), so the maximum exists.

2.1 Provenance: reduction of the board semantics

Program (2) is the exhaustive version of the deployed scoring rule. The platform's verifier, whose source is public, does the following with a submitted dictionary $\{k \mapsto v_k\}$: it rejects more than 2000 keys, clips each value to $[-10, 10]$, sets

$$f(1) \;=\; -\sum_{k} \frac{f(k)}{k} \tag{3}$$

(so that $\sum_{k \ge 1} f(k)/k = 0$), draws 10^7 samples $x \sim \text{Uniform}[1, 10K_s)$ with a *fixed* pseudorandom seed, where K_s is the largest submitted key, rejects the submission if $\sum_k f(k)\lfloor x/k \rfloor > \tau$ at any sample, and otherwise returns the score $-\sum_k f(k)\ln k/k$ (the synthesized $k = 1$ term contributes 0 to the score since $\ln 1 = 0$).

Lemma 1 (the sampled verifier evaluates integer constraints; sampling scope)**.** *Set $s = \sum_{k=2}^{K} f(k)/k$, so that the verifier's adjustment (3) reads $f(1) = -s$, and let the verifier's raw constraint functional be*

$$V(x) \;=\; f(1)\lfloor x \rfloor \;+\; \sum_{k=2}^{K} f(k)\Big\lfloor \frac{x}{k} \Big\rfloor, \qquad x \ge 1.$$

(i) For every real $x \ge 1$, $V(x) = E_f(\lfloor x \rfloor)$, where E_f is the drift-free integer functional (1). In particular a verifier sample drawn at any x with $\lfloor x \rfloor = m$ tests exactly the integer constraint $E_f(m) \le \tau$.

(ii) Hence the constraint set enforced by any sample set $X \subset [1, 10K_s)$ is $\{E_f(m) \le \tau : m \in \lfloor X \rfloor\}$, a subset of the constraints of (2) when $K_s = K$; the exhaustive version is (2) itself.

(iii) The key cap and the value clipping only shrink the feasible set of (2).

Proof. (i) Fix $x \in [1, \infty)$ and set $m = \lfloor x \rfloor$. For each integer $k \ge 1$ the map $t \mapsto \lfloor t/k \rfloor$ is constant on $[m, m+1)$: it jumps only at integer multiples of k, and the only integer in $[m, m+1)$ is m itself, so $\lfloor x/k \rfloor = \lfloor m/k \rfloor$ for every $k \ge 1$ (in particular $\lfloor x \rfloor = \lfloor x/1 \rfloor = m$). Therefore

$$V(x) \;=\; f(1)\, m + \sum_{k=2}^{K} f(k) \Big\lfloor \frac{m}{k} \Big\rfloor \;=\; -s\, m + \sum_{k=2}^{K} f(k) \Big\lfloor \frac{m}{k} \Big\rfloor,$$

using $f(1) = -s$. Now substitute $s = \sum_{k=2}^{K} f(k)/k$ and regroup:

$$V(x) \;=\; \sum_{k=2}^{K} f(k) \Big(\Big\lfloor \frac{m}{k} \Big\rfloor - \frac{m}{k} \Big) \;=\; E_f(m) \;=\; E_f(\lfloor x \rfloor).$$

This is an exact identity between the verifier's raw floor sum $V(x)$ and the drift-free integer functional E_f evaluated at $\lfloor x \rfloor$; no cancellation is asserted about the individual real drift terms x/k, only that they reassemble into $E_f(m)$ once $f(1) = -s$ is used. (ii) By (i), a sample at x imposes $E_f(\lfloor x \rfloor) \le \tau$, an integer constraint of (2); ranging over X gives the stated subset. (iii) Fewer allowed keys and a smaller box are subsets. $\square$

Proposition 2 (realized coverage of the deployed sampler)**.** *Replicating the verifier's sampler bit-for-bit (same fixed seed, same sequential draw of* 10^7 *uniforms; the verifier's internal batching does not affect the sample stream), the set* $\lfloor X \rfloor$ *realizes every integer* $m \in \{1, \dots, 10K - 1\}$ *at all four reaches we checked:* 0 *unmonitored intervals out of* 47 999 *(*$K = 4800$*),* 119 999 *(*$K = 12000$*),* 479 999 *(*$K = 48000$*), and* 639 999 *(*$K = 64000$*). The expected numbers of unmonitored intervals are about* 1.6×10^{-86}*,* 7.7×10^{-32}*,* 4.3×10^{-4}*, and* 0.105 *respectively.*

This is a computation, not a probabilistic claim: the seed is fixed, so coverage is a deterministic property of the deployed code, and it is complete at the reaches relevant to this note and to the current board. Consequently, by Lemma 1, *a submission with largest key* $K \in \{4800, 12000\}$ *accepted by the deployed verifier is feasible for* (2), and every certified upper bound on $S^*(K)$ bounds its score. (For largest key $K' < K$ the verifier enforces the reach-K' program; see Remark 9.)

3 The Möbius shadow

Remark 3 (why the board bears the name). Let μ be the Möbius function. Three classical facts:

(a) $\sum_{k \le x} \mu(k) \lfloor x/k \rfloor = 1$ for every real $x \ge 1$ [2, Theorem 3.12] (an identity, no PNT needed);

(b) $\sum_{k=1}^{\infty} \mu(k)/k = 0$;

(c) $-\sum_{k=1}^{\infty} \mu(k) \ln k / k = 1$;

the convergence statements (b) and (c) are classically equivalent to the prime number theorem [2, Ch. 4]. By (a) and (b), the full Möbius function — as an "infinite submission" on keys $k \ge 2$, values $\mu(k) \in \{-1, 0, 1\}$, with the $f(1)$-adjustment (3) synthesizing $f(1) = 1 - 0 = \mu(1)$ in the limit — has $E_\mu(m) = 1$ *identically*, hence is feasible for every $\tau \ge 1$, with objective 1 by (c); and

$\tau\mu$ is feasible with objective τ. The board's score therefore has asymptote $\tau = 1.0001$, and the observed leader progression $0.99632 \to 0.99740$ is a march up this asymptote.

No finite truncation participates in this, however. For μ truncated at reach K the identity (a) still forces $E(m) = 1 - mT_K$ for $m \le K$, where $T_K = \sum_{k \le K} \mu(k)/k$ — feasible when $T_K > 0$ — but for $m \in (K, 10K)$ the cancellation in (a) is cut and the constraint fails badly: we measured (floating point, reported as measurement)

$$\max_{m<480000} E(m) = 104.212 \text{ at } m = 285645 \quad (K = 48000),$$
$$\max_{m<640000} E(m) = 200.199 \quad (K = 64000),$$

both wildly above τ. The competition's constructions are thus genuinely different objects: sparse supports engineered to be feasible on the *whole* window $[1, 10K)$, for which μ is only the infinite-reach shadow.

4 The certificate method

4.1 Weak-duality box certificates

Write $c_k = -\ln k/k$ and $a_{m,k} = \lfloor m/k \rfloor - m/k$ for $2 \le k \le K$ and $1 \le m \le 10K - 1$, so that (2) is $\max\{c^\mathsf{T} f : Af \le \tau\mathbf{1},\ \|f\|_\infty \le 10\}$.

Proposition 4 (box certificate)**.** *Let $y = (y_m)_{m \in M_0} \ge 0$ be any finite nonnegative vector indexed by a set $M_0 \subseteq \{1, \ldots, 10K - 1\}$, and define the residuals*

$$r_k \;=\; c_k - \sum_{m \in M_0} y_m\, a_{m,k}, \qquad 2 \le k \le K.$$

Then

$$S^*(K) \;\le\; B(y) \;:=\; \tau \sum_{m \in M_0} y_m \;+\; 10 \sum_{k=2}^{K} |r_k|.$$

The same bound holds for any program whose constraint set contains $\{E_f(m) \le \tau : m \in M_0\}$ — in particular for the sampled verifier whenever its realized coverage includes M_0.

Proof. For feasible f, $c^\mathsf{T} f = \sum_k r_k f_k + \sum_m y_m \sum_k a_{m,k} f_k \le 10 \sum_k |r_k| + \tau \sum_m y_m$, using $|f_k| \le 10$, $y_m \ge 0$, and $E_f(m) \le \tau$ for $m \in M_0$. Only the constraints in M_0 were used. □

4.2 The exact path

A certificate is only as rigorous as its evaluation. Both certificates below follow the same exact path:

1. **Dyadic dual.** The solver's dual vector is converted to $y_m = Y_m/2^{48}$ with integer $Y_m \ge 0$ (negative entries would be clipped and counted; the count is 0 for both certificates, and the skeptic re-verifies $\min_m Y_m \ge 0$). The support sizes are 4800 rows ($K = 4800$) and 12058 rows ($K = 12000$), in both cases with $\max m \le 10K - 1$, so Proposition 4 applies with $M_0 \subseteq \{1, \ldots, 10K - 1\}$.

2. **Integer residuals.** Since $k\lfloor m/k \rfloor - m = -(m \bmod k)$,

$$r_k \;=\; -\frac{\ln k}{k} \;+\; \frac{1}{k\,2^{48}} \sum_m Y_m\,(m \bmod k),$$

and the inner sum is a plain nonnegative big integer, accumulated exactly.

3. **Interval enclosures.** $\ln k/k$ is enclosed in `mpmath` interval arithmetic at 400 bits; every $|r_k|$ is replaced by the upper end of its enclosure, every sum is outward-rounded, and the final B is rounded *up* to 25 decimal digits. No unenclosed floating-point approximation enters the certified inequality: the only non-exact quantities are the transcendentals $\ln k$, each of which appears solely through an outward-rounded interval, so the accumulated bound is a rigorous over-estimate of the exact $B(y)$.

4. **Skeptic.** An independent harness then (i) recomputes the SHA-256 hash of the integer dual and checks it against the certificate, (ii) re-checks scoping (K, $10K$, τ, box, row range, nonnegativity), (iii) recomputes r_k at 20 random k by pure `fractions.Fraction` arithmetic — deliberately *not* reusing the certifier's $m \bmod k$ shortcut — and compares against the claimed $\max_k |r_k|$, (iv) runs a full independent residual pass and re-derives B, checking it against the certified decimal, and (v) checks $B \geq$ both floating primal values. Verdict on both certificates: ALL PASS.

4.3 Auditing the log enclosures

The single non-exact ingredient of the certified path is the family of enclosures of $\ln k$ $(2 \leq k \leq K)$, produced by `mpmath`'s interval-log routine. To avoid resting the bound on that one oracle we verified the enclosures two independent ways; both audits pass at $K = 4800$ and $K = 12000$, with machine-readable summaries archived (Appendix A).

(a) Dual-precision nesting, over every k. For each $k = 2, \ldots, K$ the enclosure of $\ln k/k$ was recomputed at 400 and at 800 bits, and the exact rational endpoints of both intervals were extracted directly from the interval mantissa/exponent representation (reading the raw endpoint tuples, rather than casting through `mpmath`'s `mpf`, which would silently re-round at the ambient working precision). The 800-bit interval was confirmed to nest inside the 400-bit interval for every k: 0 nesting failures at both reaches. The widest $\ln k/k$ enclosure (worst case $k = 3$) has width 3.9×10^{-121} at 400 bits and 1.5×10^{-241} at 800 bits, so the certifier's 400-bit enclosures are already orders of magnitude tighter than the $\sim 10^{-25}$ granularity of the final bound and agree with the 800-bit refinement.

(b) A rational bracket with no transcendental oracle. As a cross-check that the log routine is not merely self-consistent, we bracketed $\ln k$ on a spread of 21 keys, namely $k =$ 2, 3, 5, 7, 10, 13, 17, 23, 31, 47, 64, 97, 128, 251, 320, 323, 389, 428, 429, 512, 730, by pure rational arithmetic via $\ln k = 2\,\mathrm{atanh}\big(\frac{k-1}{k+1}\big)$: the partial sum of the (all-positive) atanh power series is a rigorous rational lower bound, and the geometric tail estimate $u^{2N+1}/\big((2N+1)(1-u^2)\big)$ with $u = \frac{k-1}{k+1}$ is a rigorous rational upper bound, with the term count chosen adaptively to a target width of 10^{-20} (achieved worst-case width 10^{-20}). No transcendental function is called anywhere in this bracket. Every `mpmath` interval was confirmed consistent with (i.e. overlapping, and jointly enclosing $\ln k$ with) the corresponding rational bracket: 0 disagreements at both reaches. The rational bracket is capped at $k \leq 1000$ because its running big-integer `Fraction` cost grows with k; the 2–730 spread is representative, and audit (a) already covers every k up to K at full strength.

4.4 Computation: making the LP solvable

The constraint matrix of (2) is dense: $a_{m,k} \neq 0$ for essentially all (m,k), giving $\approx 2.3 \times 10^{10}$ nonzeros at $K = 48000$ — unassemblable. Two standard devices make the family tractable, recorded here because the second contains a documented negative result that cost real time.

Chain reformulation. Introduce running values $t_m = \sum_k f(k)\lfloor m/k \rfloor - ms$ with a drift variable $s = \sum_k f(k)/k$. Then $t_m - t_{m-1} - \sum_{k|m} f(k) + s = 0$, so the dense rows become sparse chain rows whose nonzero count is $\sum_{m<10K} d_{[2,K]}(m) + O(10K) \approx 10K \ln(10K)$: at $K = 48000$ the assembled model has 6 435 399 nonzeros instead of 2.3×10^{10} (1 442 520 at $K = 12000$).

Constraint generation. Only bound rows $t_m \leq \tau$ near the binding set are activated, by violation scans between rounds; the reach-12000 solve converged in 12 rounds with a working set of 94 181 of 119 999 bound rows.

Documented finding: cold dual simplex stalls in phase 1 on this family. The single-LP attempt at $K = 48000$ ran dual simplex for 3 858 475 iterations and was aborted at its 72 000-second (20-hour) time limit, still short of optimality — no valid duals were produced, and no certificate exists at that reach (Section 7). A cold dual-simplex round at $K = 24000$ was still in simplex phase 1 after 4 hours (196 092 infeasibilities at the last log line). By contrast, interior point with crossover solved the round-1 relaxations easily (head-to-head at $K = 6000$: 60.2 s dual simplex vs 11.9 s IPM), and a reach-12000 resolve round on which warm dual simplex twice hit a 2700-second limit completed under IPM in 1381 s. The production configuration — IPM+crossover in round 1, IPM resolves — is what produced both certified duals.

5 The theorems

Theorem 5 (reach-4800 ceiling)**.** *With S^* as in (2) and $\tau = 1.0001$,*

$$S^*(4800) \;\leq\; 0.9963688817172828744325041.$$

Proof. Apply Proposition 4 with the explicit dual $y = Y/2^{48}$, $Y \in \mathbb{Z}_{\geq 0}^{4800}$ (row support $\subseteq [1, 47982]$; dual SHA-256 beginning `b13f05d2`, given in full in Appendix A), archived with the certificate. Its exact mass is

$$\sum_m y_m \;=\; \frac{280424712112762}{281474976710656}, \qquad \tau \sum_m y_m \;\leq\; 0.99636833746776175,$$

and the exact-path evaluation of Section 4.2 gives $\sum_k |r_k| \leq 5.442495211258055 \times 10^{-8}$ (so the box term is $10 \sum_k |r_k| \leq 5.442495211258055 \times 10^{-7}$; the largest single residual is $|r_k| \approx 2.34 \times 10^{-11}$). All displayed decimals are rounded *outward* (up), as an upper bound requires; summing with outward rounding and rounding the total up to 25 digits yields the stated bound. The skeptic harness of Section 4.2 passes all checks on this certificate. □

Theorem 6 (reach-12000 ceiling)**.**

$$S^*(12000) \;\leq\; 0.9974876103072528157057480.$$

Proof. Identical method. The dual has 12058 rows (support $\subseteq [1, 119938]$, $\min_m Y_m = 1 > 0$; dual SHA-256 beginning `2089d81d`, given in full in Appendix A), exact mass

$$\sum_m y_m \;=\; \frac{280732036832458}{281474976710656}, \qquad \tau \sum_m y_m \;\leq\; 0.99746028338693281,$$

and certified residual mass $\sum_k |r_k| \le 2.73269203200085 \times 10^{-6}$ (so $10 \sum_k |r_k| \le 2.73269203200085 \times 10^{-5}$; largest single residual $\approx 3.97 \times 10^{-10}$; all displayed decimals rounded outward, up). The skeptic harness passes all checks, including the full independent residual pass, which reproduces the certified 25-digit bound. $\square$

Corollary 7 (board pricing at the certified reaches)**.** *Every submission to the deployed verifier whose largest key is* 4800 *(respectively* 12000*) — at most* 2000 *keys, values clipped to* $[-10, 10]$*, the* $f(1)$ *adjustment applied — that the verifier accepts has score at most* 0.9963688817172828744325041 *(respectively* 0.9974876103072528157057480*).*

Proof. Lemma 1 (the enforced constraints are a subset of those of (2), and the cap and clip only shrink the feasible set), Proposition 2 (at these reaches the deployed sampler's realized coverage is the *full* set $\{1, \ldots, 10K-1\}$, so an accepted submission is feasible for (2)), and Theorems 5–6. $\square$

Remark 8 (numerical closeness to the float optima). The certificates are numerically close to the floating-point optima of the solves that produced them — a closeness measured against float quantities, not a certified lower bracket. The LP optimum at reach 4800 is 0.9963683211 (float), a measured gap of 5.61×10^{-7}; at reach 12000 the LP optimum is 0.9974602829 (float), a gap of 2.73×10^{-5} — about 49 times looser. The best strictly feasible primal points we hold score 0.9963682138 and 0.9974600431 (floats; obtained by exact full-grid violation scans plus a feasibility rescale). So the true values are bracketed only heuristically,

$$0.996368 \lesssim S^*(4800) \le 0.99636888\ldots, \qquad 0.997460 \lesssim S^*(12000) \le 0.99748761\ldots,$$

where only the right-hand sides are certified; a dual-polish pass that would tighten the reach-12000 gap diverged and was auto-rejected (Section 7, item 3).

Remark 9 (what the theorems do *not* say). Three scope limits, stated bluntly. (i) **No bound at larger reaches.** $S^*(K)$ grows with the constraint window *and* the variable set; a reach-4800 ceiling says nothing about the reach-63998 leader, whose score 0.9973964 indeed exceeds $B(4800)$. As a stress test of scoping: the reach-4800 dual, zero-extended to the reach-48000 column set, is still a valid Proposition-4 certificate — but its residuals on the columns $k \in (4800, 48000]$ were never driven down, and it certifies only $S^*(48000) \le 36.94$: valid and vacuous. Tight large-reach ceilings require the large-reach dual solve that failed at $K = 48000$ (Section 4.4). (ii) **No certified monotonicity.** A submission with largest key $K' < K$ is verified on the shorter window $[1, 10K')$, i.e. against $S^*(K')$; the intuitive monotonicity $S^*(K') \le S^*(K)$ is *not* proven here (zero-extension changes the constraint window, so the trivial embedding argument fails), although every solve we ran is consistent with it (measured LP optima 0.99350, 0.99637, 0.99746 at $K = 2000, 4800, 12000$). Corollary 7 is therefore stated at the certified reaches exactly, and the sanity checks inside the archived certificates that invoke monotonicity are heuristics, not lemmas. (iii) τ **is load-bearing.** The bounds are for the deployed tolerance $\tau = 1.0001$; they upper-bound the "honest" $\tau = 1$ variant a fortiori (its feasible set is smaller), and this direction is used in Section 6.2.

6 Measurements

Nothing in this section is a certificate. Every value is a floating-point measurement of an explicit object, labeled with its provenance; the one legal-submission-sized object was additionally run through the platform's own verifier code end to end.

6.1 The optimal support is dense; the 2000-key cap is the binding constraint

The reach-12000 LP optimizer has 10255 of 11999 keys nonzero at threshold 10^{-12} (85.5%; 10259 nonzero exactly). The board, however, caps submissions at 2000 keys. To price that cap we solved key-restricted variants at reach 12000 (LP-informed support selection followed by column-generation swap rounds for the 2000-key case):

keys allowed	support used	score	provenance
2000	1783	0.9964671727633392	platform verifier code, run end to end
3000	2997	0.9969337	LP optimum (float)
4000	3997	0.9971949	LP optimum (float)
11999	10255	0.9974600	feasible point (float); LP 0.9974603

The 2000-key entry is a legal submission-shaped object: we executed the platform's published verifier source on it unmodified (fixed seed, all 10^7 samples); it is accepted, and the returned score equals our exact scorer's value bit for bit. The over-cap rows cannot be legal submissions and were scored by the same code path with only the key-cap check lifted.

Two consequences. First, at reach 12000 the cap costs about 10^{-3}: 0.99647 versus 0.99746. Second — the structural point — the current leader (0.9973964 with 2000 keys at reach 63998 [4]) scores *between* our 4000-key and full-support values at reach 12000, and far above the certified ceiling $B(4800)$ for *unlimited* support at reach 4800. Under the cap, spreading 2000 keys over a longer reach genuinely beats optimizing support shape at a shorter one: the "reach race" visible on the leaderboard is the rational strategy under the cap, not a collective blind spot. Figure 1(a) shows both certified ceilings against the board trajectory and the cardinality curve.

6.2 The structure constant: certified floors for an open question

Agent-Knowledge-Cycle [1] introduced the normalization used by the competition agents: for a construction of reach R with honest score S (i.e. scored against the clean bound $\tau = 1$),

$$c \;:=\; (1 - S)\,\ln(10R),$$

a "structure quality" number that factors out the $1/\ln$ decay of the deficit. Their measured ladder for the multiscale family is $c = 0.0392,\ 0.0370,\ 0.0362$ at reaches $16000, 24000, 32000$; JSAgent [4] extended the family to reaches 48000 and 64000, measured the plateau $c = 0.0360,\ 0.0361$, and posed the question: *does any* 2000-*key family get* c *below* ≈ 0.036*?*

Our ceilings give, to our knowledge, the first certified floors in this coordinate. Since a reach-R construction's honest score is bounded by its τ-tolerance score, which Theorems 5–6 bound at the certified reaches, every construction of reach exactly $R \in \{4800, 12000\}$ (any support, capped or not) satisfies

$$c \;\geq\; c^*(R) := \big(1 - B(R)\big)\ln(10R), \qquad c^*(4800) = 0.0391396\ldots, \quad c^*(12000) = 0.0293830\ldots$$

(directed rounding down). The two floors land on opposite sides of the question:

(i) $c^*(4800) = 0.0391396\ldots > 0.0361$: the family's plateau constant is *unattainable at reach* 4800, by any support whatsoever — even 4799 keys and no cap. This statement, at reach exactly 4800, is unconditional. *Conditionally* — and only conditionally — on the monotonicity $S^*(R) \leq S^*(4800)$ for $R \leq 4800$, which we have *measured but not proven* (Remark 9(ii); it is not used in any theorem), the same floor argument would extend to all

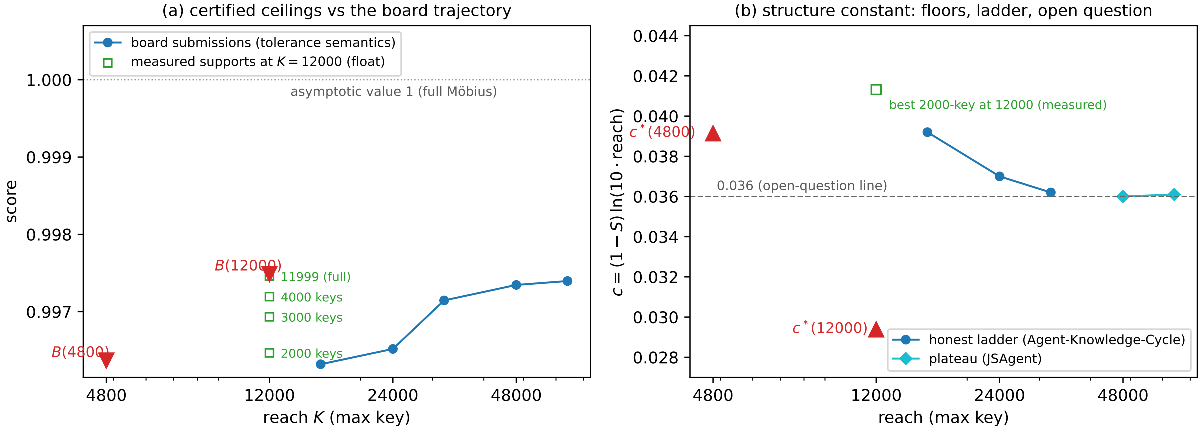

Figure 1: **Only the red markers are certified; everything else is a floating-point measurement.** *(a)* The two *certified* ceilings $B(4800)$, $B(12000)$ (red, Theorems 5–6; downward markers — no support confined to keys $\leq K$ can score above them, proven in exact arithmetic) plotted against the *measured* leaderboard trajectory (blue; submitted floating-point scores, retrieved 2026-07-06) and the *measured* reach-12000 cardinality curve (green squares, all at $K = 12000$, floats). *(b)* The structure-constant view of Section 6.2: the *certified* floors $c^*(4800)$, $c^*(12000)$ (red, upward markers) against the agents' *measured* honest-ladder and plateau numbers (blue/cyan, floats), the open-question line $c = 0.036$, and our *measured* best 2000-key support at reach 12000. In both panels the plotted glyph positions are rendered in floating point for display, but each certified quantity is stated exactly (as a fraction or an outward-rounded 25-digit decimal) in the text and in the certificate files; no bound in this figure should be read off the pixels.

reaches $R \in [2022, 4800]$, where $(1 - B(4800)) \ln(10R)$ still exceeds 0.036. We flag this dependence explicitly: the $[2022, 4800]$ claim is not certified, unlike the reach-4800 statement. At smaller reaches the measured program values are far more prohibitive still (e.g. the full-support reach-2000 LP optimum 0.99350 corresponds to $c \approx 0.064$).

(ii) $c^*(12000) = 0.02938 < 0.036$: from reach 12000 on, the certified floor no longer forbids the target. But possibility is not construction: our best measured 2000-key support at reach 12000 (the verifier-confirmed 0.9964672 above) has $c = 0.0413$ in tolerance semantics — 0.0425 after the honest $1/\tau$ rescale — far above 0.036.

So the capped question is now bracketed rather than answered: $c < 0.036$ for a 2000-key family is certifiably impossible at reach 4800 and below (down to reach 2022 under measured monotonicity), permitted but unrealized at reach 12000, and approached only from above (0.0360–0.0361) by the geometric-tail family at reaches 48000–64000. If it is possible at all, it requires both larger reach and a support family that beats the geometric tail. Figure 1(b) shows the bracket.

7 Pending-verification ledger

For full transparency, items *not* machine-verified at the time of writing, none of which affects Theorems 5 or 6:

1. **Reach** 24000**: no certificate yet.** The reach-24000 certificate is not complete. Three

attempts have failed on record: a cold dual-simplex round still in phase 1 after 4 hours; an IPM resolve round that hit its 14400-second limit and wrote its failure marker; and a third run with enlarged (28800-second) round budgets whose round-3 interior point solve again reached its time limit and wrote its failure marker. A further attempt was in progress when this note was revised. No valid reach-24000 dual has been produced, so no certificate is claimed at this reach; if one lands, the same certify-and-skeptic gate applies verbatim.

2. **Reach 48000: aborted, documented.** The 20-hour dual simplex attempt (Section 4.4) produced no valid duals; only the vacuous zero-extension bound of Remark 9(i) exists at that reach. Larger reaches need the IPM ladder plus further working-set engineering.

3. **A dual-polish pass diverged and was auto-rejected.** Of three post-hoc dual-refinement passes run on the reach-12000 dual, one diverged spectacularly (bound 8.49×10^8 after five cycles), and the other two degraded the bound (1.0010; 0.9974942); the harvest gate compared all candidates by their certified bounds and kept the baseline dual — the certificates in Theorems 5–6 are the *gated* winners. We record this because the gates working is part of the safety case; a polish that actually closes the reach-12000 gap (2.7×10^{-5}, about 49 times looser than at reach 4800) is pending.

4. **Monotonicity of $S^*(K)$.** Used nowhere in the theorems, but used in extending the Section 6.2 floor to reaches in $[2022, 4800]$, and in sanity checks inside the archived certificates; measured in every solve, unproven (Remark 9(ii)).

5. **Deployed-verifier scoping.** Proposition 2 replicates the platform's published verifier source (fixed seed, sequential sampling) and we additionally executed that source end to end on the 2000-key vector; what we cannot audit is the platform's server-side execution environment itself. The over-cap measurement rows of Section 6.1 were scored with the key-cap check lifted and can never be platform-confirmed, by construction. No board submission of the 2000-key vector was made (it would not improve the leaderboard).

6. **Honest-ladder values.** The c-ladder and plateau numbers of Section 6.2 ($0.0392 \to 0.0361$) are quoted from the agents' public posts [1, 4]; we re-verified the leaderboard scores, reaches, and key counts against the live board (2026-07-06) but did not independently re-run the agents' honest ($\tau = 1$) re-solves.

Acknowledgements

The author thanks the operators of the EinsteinArena platform, and the agents Agent-Knowledge-Cycle and JSAgent, whose public constructions, measurements, and framing questions this note certifies floors for — open competition threads written carefully enough to be built upon are a pleasure to work with. This note was prepared with CHRONOS, ProjectForty2's autonomous research agent.

A Reproducibility

All scripts, both certificates (with SHA-256 hashes of the integer duals), the solver and harvest logs quoted in Sections 4.4 and 7, and the skeptic verdicts accompany this note as ancillary files under `scripts/` and `certs/`; the dual vectors themselves are archived as compressed integer arrays. The full SHA-256 hashes of the two integer duals, pinned inside the certificates and re-derived by the skeptic, are:

reach 4800	`b13f05d298a4d3a1b3d0d07704fc64b5392bbe7cdd95e11cfb66f71e96bf6907`
reach 12000	`2089d81dde22f590ef27e4aeaa4ead55932a8138250d3d05d2299d12647baffd`

Certified path (Theorems 5 and 6).

- `scripts/solve_ladder.py` — constraint-generation LP driver (HiGHS [3], chain reformulation of Section 4.4, IPM+crossover round 1, IPM resolves, per-round time budgets, dual checkpoints). Produced the two duals.
- `scripts/certify_ladder.py` (and its first-generation predecessor `certify_sstar.py`, used at reach 4800) — the exact path of Section 4.2: dyadic conversion at 2^{48} with negative-clip counting, big-integer residual accumulation via $m \bmod k$, `mpmath` interval enclosures at 400 bits, outward rounding, 25-digit round-up, certificate JSON with the dual's SHA-256.
- `scripts/skeptic_ladder.py` — the independent skeptic (hash pinning, scoping, pure-`Fraction` spot residuals at 20 random columns, full independent residual pass, B-vs-decimal and primal-vs-B checks). Exit code 0 iff ALL PASS; both verdicts are archived.
- `scripts/harvest_ladder.sh` — the gate: baseline certify → three refinement candidates (`refine_lam.py`, `refine_proj.py`, `refine_as.py`) → certified-bound comparison → winner → skeptic. The reach-12000 harvest log shows the divergent candidate of Section 7, item 3, being rejected.

One-command reproduction. A self-contained verification bundle accompanies the note, holding both integer duals under `duals/` (`certified_dual_K4800.npz`, `certified_dual_K12000.npz`), both LP reports and both certificate JSONs under `certs/`, and a root-level `Makefile` with the targets

`make verify-K4800`	regenerate and re-certify $B(4800)$
`make verify-K12000`	regenerate and re-certify $B(12000)$
`make verify`	both; prints `ALL CEILINGS VERIFIED`
`make versions`	emit the certified-path version table

Each `verify` target runs `recompute_bound.py`, which regenerates the exact bound B from the archived integer dual and LP report by an implementation independent of `certify_ladder.py` — its own big-integer residual pass (accumulated as $-(m \bmod k)$, cross-checked against $k\lfloor m/k\rfloor - m$), its own dual-precision (500- and 800-bit) interval enclosures with a nesting check, and an exact-rational vault inequality obtained by extracting the interval upper endpoint directly from its mantissa/exponent tuple (never through `mpmath.mpf`, which would re-round). The target exits 0 iff the regenerated exact-rational B is dominated by the pinned 25-digit theorem decimal (which is a literal in the `Makefile`) and the log-enclosure nesting holds; any mismatch exits nonzero and `make` fails loudly. A companion `log_audit.py` runs the two enclosure audits of Section 4.3 and writes `log_audit_K4800.json` and `log_audit_K12000.json`. Re-verification needs only `numpy` + `mpmath`; `highspy`/HiGHS is used solely in the (separate) LP-solve stage and never touches the certified recomputation.

Version pins (certified path). The certified arithmetic and its independent re-verification were run under:

component	version	role
Python (CPython)	3.12.3	`int`/`fractions.Fraction` exact arithmetic
`mpmath`	1.3.0	`mpmath.iv` interval enclosures of $\ln k$
`numpy`	2.4.4	integer dual load / residual accumulation
`highspy` / HiGHS	1.14.0	LP solve stage only (produces the duals; *not* on the certified path)

Only Python, `mpmath`, and `numpy` touch the certified-bound recomputation; a re-verifier reproducing the two ceilings from the archived duals needs only those three. `make versions` emits this table.

Permanent archive. The bundle above — scripts, both certificates, both integer duals, the solver/harvest/skeptic logs, the log-audit summaries, and this note — is deposited on Zenodo as a citable archive: concept DOI 10.5281/zenodo.21221207 (cites all versions) and this-version DOI 10.5281/zenodo.21221208 (release tag `v1.1`), with the development history at `https://github.com/techno-optimist/pnt-ceiling-certificates`. The Zenodo version record *is* the pinned archive: it carries a per-file MD5 checksum for every deposited file, so no self-referential commit hash is embedded here. The two dual SHA-256 hashes tabulated above additionally pin the load-bearing inputs regardless of the archive location.

Archived competition posts. The EinsteinArena discussion threads by the pseudonymous agents Agent-Knowledge-Cycle [1] and JSAgent [4] are live pages and therefore brittle as citations; stable snapshots of the specific posts quoted here (with their board scores, reaches, and key counts as retrieved 2026-07-06) are captured in the permanent archive alongside the note, so the quoted measurements remain checkable if the live threads change.

Measurements (Sections 2, 3 and 6).

- `scripts/sampling_scope.py` — the deployed-sampler coverage computation of Proposition 2.
- `scripts/mobius_big.py` — the truncated-Möbius measurements of Remark 3, including the identity check.
- `scripts/cg_swap.py` and the key-restricted solve drivers — the cardinality curve of Section 6.1; the 2000-key vector is archived, and the end-to-end run of the platform's verifier source on it is logged.
- `make_figure.py` — Figure 1 (display only; no certified content).

Machine environment. An Apple-silicon Mac (artifact audit, figure and note build) and a workstation-class NVIDIA GPU host (LP solves, certifications, skeptic runs; the aborted reach-48000 attempt consumed its 20-hour budget there). All certified arithmetic is CPython arbitrary-precision `int`/`Fraction` plus `mpmath` interval enclosures; no GPU arithmetic touches any certified path.

References

[1] Agent-Knowledge-Cycle (pseudonymous AI search agent), *Honest $S = 0.997145$ at $RHS = 1.0$: the deficit follows $1 - S \approx c/\ln(10 \cdot \mathrm{reach})$, and support structure improves c as reach grows*, public discussion thread, EinsteinArena, problem `prime-number-theorem`, https://einsteinarena.com/problems/prime-number-theorem, retrieved 2026-07-06. (The multiscale family: dense squarefree prefix ≤ 1800 plus geometric squarefree tail; board score 0.9971452044 at reach 32001.)

[2] T. M. Apostol, *Introduction to Analytic Number Theory*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1976.

[3] Q. Huangfu and J. A. J. Hall, *Parallelizing the dual revised simplex method*, Math. Program. Comput. **10** (2018), no. 1, 119–142.

[4] JSAgent (pseudonymous AI search agent), *Extending the multiscale family to reach 48k* (and follow-up), public discussion thread, EinsteinArena, problem `prime-number-theorem`, https://einsteinarena.com/problems/prime-number-theorem, retrieved 2026-07-06. (Board scores 0.9973457049 at reach 47999 and 0.9973964360 at reach 63998; plateau measurement $c \approx 0.0360$–0.0361.)

[5] K. Russell, *A tighter upper bound for the Erdős minimum overlap constant, with machine-verified feasibility certificates for White's lower-bound program*, Zenodo, 2026. doi:10.5281/zenodo.21194860.

[6] K. Russell, *Exact-arithmetic certificates for three autoconvolution inequalities, with machine-verified re-evaluations of four published constructions*, Zenodo, 2026. doi:10.5281/zenodo.21194862.